

Aaron Madved

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OVERVIEW

Full-stack engineer with experience across the entire delivery stack — from React/Angular frontends to Node.js/Go backends, cloud infra, and DevOps pipelines. Hands-on with Web3 development, EVM smart contracts, and DeFi protocols. Recently focused on AI tooling and multi-agent systems, including building Docker-based swarm orchestration infrastructure for autonomous Claude Code agents. Comfortable in startup environments, remote-first teams, and contractor arrangements.

WORK EXPERIENCE

Tech Lead — CGI, Pittsburgh, PA · 2024–Present

Tech lead across three concurrent client engagements spanning infrastructure, full-stack development, and delivery management.

- **Infrastructure** — Maintained and optimized Jenkins CI/CD pipelines and OpenShift deployments, managing manifests, pod scaling, and container orchestration across environments.
- **Full-Stack Engineering** — Resolved critical bugs across the entire stack, from Angular frontends through Spring Boot microservices to MongoDB and Oracle database layers.
- **Tech Lead / Delivery** — Owned release management, sprint planning, and task delegation across the dev team. Served as the primary liaison between engineering, product owners, and business stakeholders to align expectations and drive delivery.

Lead Backend Engineer — BCW, Hong Kong · 2021–2024

Led backend infrastructure and node operations across four core products at a blockchain infrastructure company.

- **StakeFi** — Architected and maintained an enterprise-grade B2B PoS staking platform supporting 15+ L1/L2 protocols. Managed full AWS infrastructure including IAM, EC2, S3, Route53, CloudFront, and ElastiCache.
- **Hashport** — Built and maintained a cross-chain bridge between Hedera and EVM networks. Designed the validator swarm architecture and developed a modular HCS consensus service integrated with PolygonSDK.
- **Arkhia** — Founding member and architect for Hedera RPC provider, enabling reliable programmatic access to the Hedera network.
- **Node Operations** — Founded and scaled a globally distributed DevOps team responsible for 24/7 node operations across multiple chains. Established the foundational systems and processes still in use today.

Software Consultant — CGI, Pittsburgh, PA · 2020–2021

Frontend consultant embedded with client teams across the USA, focused on scalable UI architecture and delivery.

- **Component Library** — Maintained and expanded a shared Angular component library for one of the largest regional banks, ensuring consistency, reusability, and scalability across

multiple applications.

- **Accessibility (A11y)** — Enforced WCAG compliance and accessible component patterns across the library, ensuring applications met federal and industry accessibility standards.
- **Micro UI Architecture** — Designed and implemented micro frontend architecture, enabling independent deployment and ownership of UI modules across distributed teams.

Full-Stack Engineer — Microsoft/CMU, Pittsburgh, PA · 2019–2020

Embedded engineer on CMU's cloud computing program, sponsored by Microsoft Azure to modernize and scale the curriculum across top-tier universities.

- **SAIL Foundation** — Led frontend and infrastructure development for the SAIL initiative, extending Professor Sadeh's cloud computing course to support multiple LMS platforms for Ivy League institutions including Harvard. Built the reverse proxy layer and core integration architecture enabling cross-institutional adoption.
- **Cloud Computing Platform** — Maintained the Django/React application used by CMU students for coursework submission and automated grading. Updated Terraform scripts for student lab environments and managed the grading pipeline.
- **TA & Curriculum Ops** — Managed a team of master's student TAs, coordinating course material creation, project scoping, and student deliverable reviews to keep the program running at scale.
- **Microsoft Azure Partnership** — Worked directly with the Azure team to beta test new cloud features within the student environment, providing structured feedback that informed Azure product development. Microsoft sponsored the engagement to drive adoption of cloud-native tooling in academia.

Software Engineer — Li-x GmbH, Hamburg, Germany · 2018–2019

- **Frontend Modernization** — Led a full rewrite of a legacy PHP/Symfony application into a modern Angular SPA, significantly improving performance, maintainability, and user experience.
- **Backend Migration** — Built the accompanying Express/Node.js backend to replace the Symfony layer, decoupling frontend and backend concerns into a clean API-driven architecture.
- **Infrastructure** — Containerized the full web stack with Docker and provisioned GCP infrastructure to support the migrated platform, establishing the foundation for scalable cloud deployment.

Software Engineer — Viewfin, Shanghai, China · 2017–2018

Early-stage engineer at a blockchain company during the height of China's ICO boom, contributing across two crypto exchanges and the Metaverse blockchain ecosystem.

- **RightBTC Exchange** — Built core frontend components and views in Angular, including candlestick charting and real-time graph data visualizations for live crypto trading.
- **SZZC Exchange** — Founding team member for a Chinese digital currency exchange, contributing to frontend architecture and Express API development from the ground up.
- **Metaverse Blockchain** — Developed key ecosystem applications including the Metaverse Light Wallet (SPV client) and Metaverse Explorer, providing users with wallet management and on-chain data visibility.
- **Context** — Worked at the epicenter of China's 2017 ICO surge, gaining firsthand experience in high-traffic crypto infrastructure before the government crackdown reshaped the market.

EDUCATION

Kent State University - BS - Application Development - 2011 - 2016

SKILLS

Languages: Javascript/Typescript, Python, Golang, Bash/Shell, Java, Perl

Dev FE: Angular, React, Next.js, SSR/CSR, SCSS/CSS, jQuery, A11y, Micro UIs, Nx

Dev BE: Nodejs/Express/Nest, Django, Springboot, Stdlib/Chi, MySQL, Postgres, MongoDB, Redis, Kafka, RabbitMQ, REST, GraphQL, gRPC

DevOps: Jenkins, Github Actions, ArgoCD, Skaffold, Docker, Ansible, Terraform, Datadog, Grafana, Prometheus

Infra: AWS(IAM, EC2, S3, Route53, CloudFront, ElastiCache), GCP(GKE, AA, CE), OpenShift, K8s, K3s, microk8s Helm, Terraform, Unix Redhat, SystemD, Cron, RPM, DEB, Busybox, Coreutils

Crypto: Solidity, Ethers.js, Web3.js, Hardhat, Foundry, The Graph, Wagmi, ERC-20, ERC-721, ERC-1155, DEX, AMMs, DeFi, L1/L2 Chains, Smart Contract Dev & Auditing, Gas Optimization, Wallet Integration

AI: Claude API, Claude Code, OpenCode, OpenRouter, MCP, RAG, Multi-Agent Orchestration, Swarm Architecture, Docker-Isolated Agent Environments, LLM-Integrated Pipelines, Prompt Engineering